

Qatar-1b

R-2808

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_130730 · created 2026-09-05T01:02:20+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2456503.9077 +/- 9.2e-05 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.186 +/- 0.038
Transit depth (Rp/Rs)^2	3.47 +/- 1.41 [%]
Orbital Inclination (inc)	81.16 +/- 2.56
Airmass coefficient 1 (a1)	1.0 +/- 0.014
Airmass coefficient 2 (a2)	-0.002 +/- 0.017
Scatter in the residuals of the lightcurve fit is	1.44 %
Best Comparison Star	#2 - [453, 221]
Optimal Aperture	3.09
Optimal Annulus	13.67
Transit Duration (day)	0.045 +/- 0.027

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.186 ± 0.038 vs 0.1463 (deviation 1.04σ)
Duration fit vs published	0.045 d vs 0.0692 d (deviation 0.9σ)
Mid-time O–C	-0.1 min
Depth SNR	4.9
Residual scatter	1.44 %

Light curve

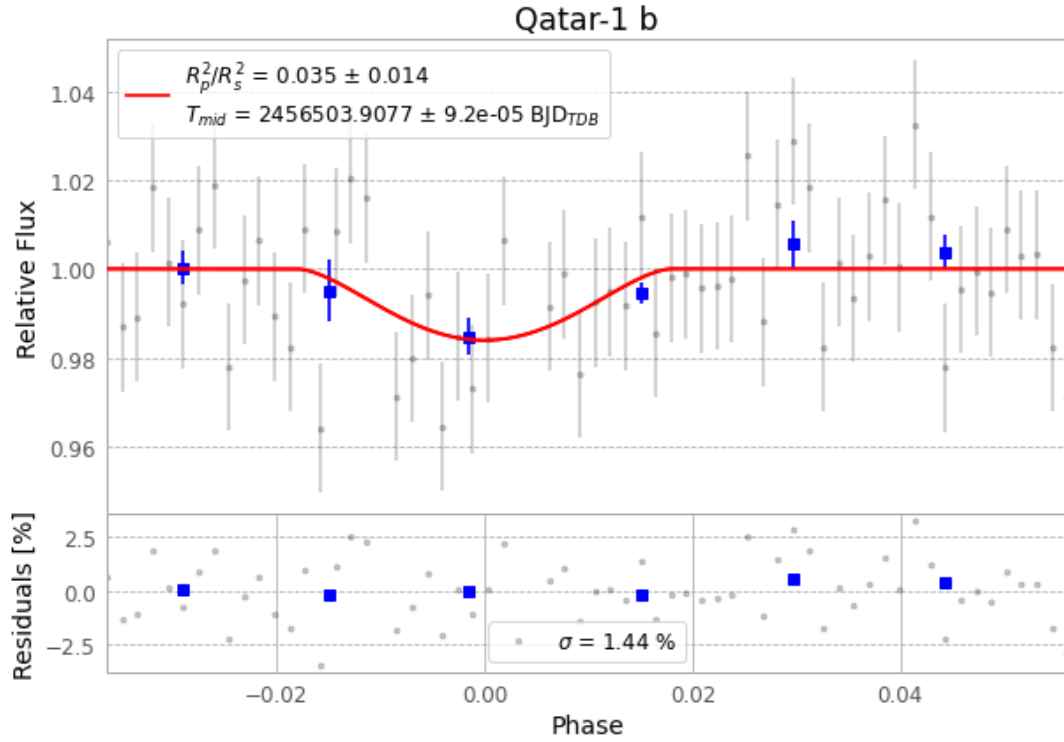


Figure 1 — FinalLightCurve_Qatar-1 b_2013-07-30.png (SHA-256 d7c5a1ffcd7f2a40...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [406, 278], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 c8ef93d4c74bb129...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

64 FITS frames (42 MB), Qatar-1130730083012.FITS → Qatar-1130730113912.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-130730.log (046e910ac93c...), FinalLightCurve_Qatar-1 b_2013-07-30.png (d7c5a1ffcd7f...), FinalLightCurve_Qatar-1 b_2013-07-30.csv (ed02f957f571...)

Compute resources

Wall time 370.0 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-09-05 01:02Z.